

MUSTAFA SHOUKAT

Senior AI Engineer, Enterprise AI Architecture and Technical Leadership

mustafashoukat.ai@gmail.com linkedin.com/in/mustafashoukat github.com/Mustafa-Shoukat1 Riyadh, Saudi Arabia

Executive Positioning Brief

Prepared for Staff AI Engineer, Teradata, 23 September 2026

This brief maps my experience to the requirements of the Staff AI Engineer role. For each area it states what the role asks for, what I have done, and the result.

1. AI platform architecture

The requirement. Design and deliver highly scalable AI products; architect for scalability, reliability and performance.

What I did. I designed ByanRAG 2.2 end to end: FastAPI services, a React admin console with 14 panels, isolated Qdrant collections per organization, a Casbin access model with five role levels checked on every endpoint, and full audit logging. The platform is packaged for on premises and air gapped deployment.

Result. Two organizations and more than 150 users share one platform with strict data isolation, inside regulated networks where cloud services are not an option.

2. Retrieval augmented generation

The requirement. Apply LLMs and retrieval to enterprise data with answers people can trust.

What I did. I built hybrid retrieval that combines dense embeddings with BM25, merges them with Reciprocal Rank Fusion and reranks with a CrossEncoder. It is tuned for Arabic first, bilingual content and streams answers with source citations. Quality is measured with Recall@5, MRR and NLI faithfulness scoring.

Result. Cited answers in 2 to 3 seconds over large document collections on self hosted open source models, with retrieval quality tracked as numbers rather than judged by eye.

3. Orchestration and agentic AI

The requirement. Experiment with and deliver agentic AI and large language model systems.

What I did. I lead multi agent workflows in LangGraph and CrewAI with tool calling and multi step planning. On Global Holman, agents pull solicitations from SAM.gov by NAICS code, read the PDFs through a RAG pipeline, analyze requirements, support the bid decision and draft the proposal over a provider agnostic LLM layer. On Whoza.ai, a voice agent answers calls and qualifies jobs for UK tradespeople. I also build with MCP: my own MCP server with Arabic tools, and MCP connected tools in my daily development workflow.

Result. Analyst work that took days is now automated, and the choice of LLM provider stays open.

4. Distributed systems

The requirement. Strong grasp of distributed system patterns, data consistency and scale.

What I did. I use Celery and Redis for background ingestion, scraping and agent jobs; async PostgreSQL for data; WebSockets and server sent events for streaming; idempotent, signature verified webhooks; and swappable vector stores behind one interface. On VacantSeek 2.0 I moved the platform onto AWS ECS with CodeBuild pipelines and S3 storage.

Result. Work keeps running when a worker fails, retries never create duplicate records, and components can be replaced without rewriting the system.

5. Production readiness

The requirement. Resolve performance bottlenecks, investigate production issues and maintain code quality.

What I did. I run services in Docker behind nginx with TLS, with CI/CD and a suite of more than 600 tests on ByanRAG. Monitoring uses Prometheus, Grafana and Loki, with LangSmith tracing and Sentry. Security covers role based access, audit logs, guardrails and prompt injection defenses. On VacantSeek 2.0 I led more than 340 merged pull requests of production hardening, including build gates on critical vulnerabilities.

Result. Problems surface on dashboards and traces before users report them. The same evaluation discipline helped take Yehia R1 to first place on the AraGen Leaderboard.

6. Technical leadership

The requirement. Mentor engineers, influence architecture decisions and work independently.

What I did. I lead and mentor the ByanRAG team of four (AI, LLM, frontend, and DevOps and QA) and own its architecture decisions, setting the standards for testing, deployment, code review and model integration. I have trained more than 150 professionals in data science and AI engineering, work directly with clients in four countries, and contribute to LangChain, RAGAS and Unstructured.

Result. The team ships to a shared standard, and I can carry a system from first design to production on my own.

Engineering foundations

Languages. Python as my primary language, with Java, TypeScript and SQL. **APIs and data.** RESTful APIs in FastAPI; data modeling in PostgreSQL, MongoDB, Redis and pgvector with SQLAlchemy and Alembic. **Cloud and containers.** AWS, GCP and Azure, with Docker in production. **AI productivity tools.** I use GitHub Copilot and MCP connected coding assistants every day.

I run containers with Docker and AWS ECS and messaging with Celery and Redis; the same patterns carry directly to Kubernetes and Kafka.

Summary

I bring proven, hands on ownership of the six areas this role centers on: platform architecture, RAG, agentic orchestration, distributed systems, production readiness and technical leadership, backed by systems that run in production today.